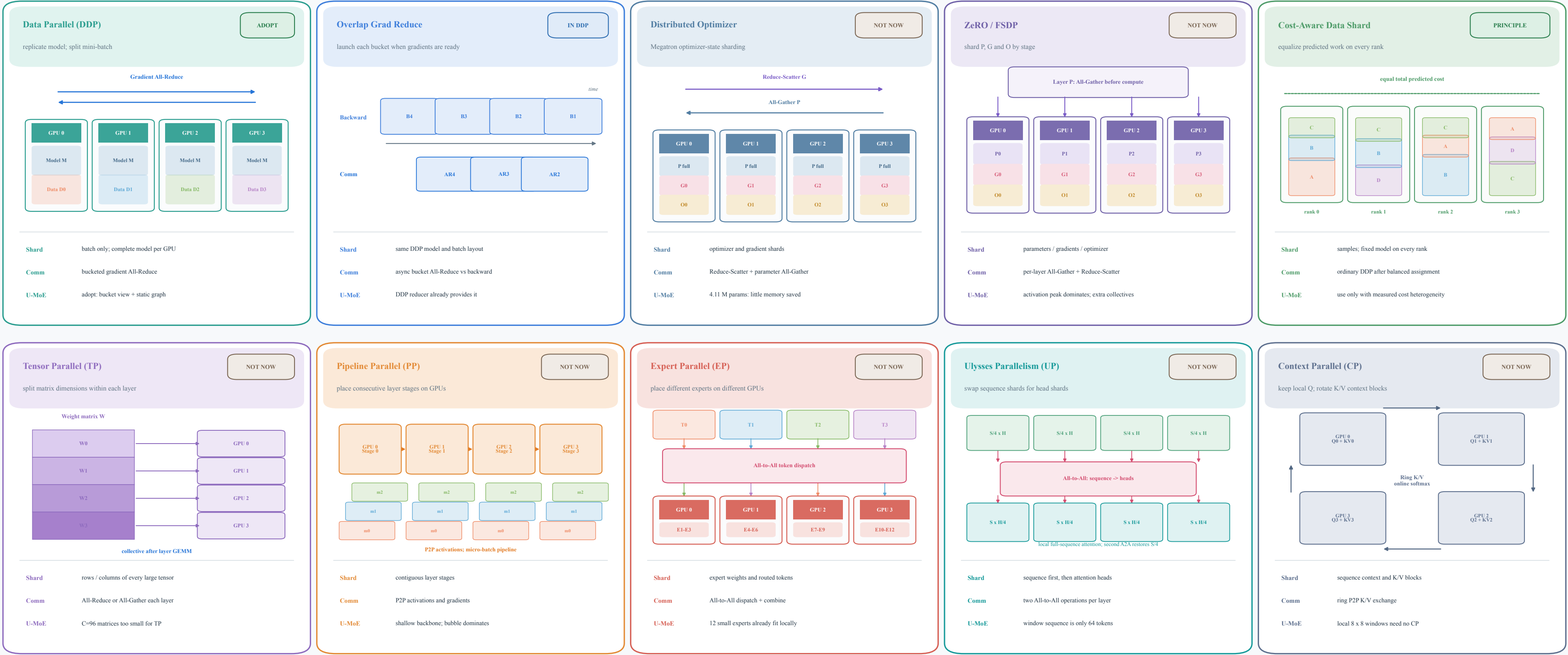


Distributed Training Mechanisms: What Is Sharded and How It Communicates

Mechanisms from Table 4-21 plus TP, PP, Ulysses Parallelism and Context Parallelism



IN DDP

Overlap Grad Reduce

launch each bucket when gradients are ready

Backward

B4

B3

B2

B1

Comm

AR4

AR3

AR2

Shard

same DDP model and batch layout

Comm

async bucket All-Reduce vs backward

U-MoE

DDP reducer already provides it

NOT NOW

Distributed Optimizer

Megatron optimizer-state sharding

Reduce-Scatter G

All-Gather P

GPU 0

P full

G0

O0

GPU 1

P full

G1

O1

GPU 2

P full

G2

O2

GPU 3

P full

G3

O3

Shard

optimizer and gradient shards

Comm

Reduce-Scatter + parameter All-Gather

U-MoE

4.11 M params: little memory saved

NOT NOW

ZeRO / FSDP

shard P, G and O by stage

Layer P: All-Gather before compute

GPU 0

P0

G0

O0

GPU 1

P1

G1

O1

GPU 2

P2

G2

O2

GPU 3

P3

G3

O3

Shard

parameters / gradients / optimizer

Comm

per-layer All-Gather + Reduce-Scatter

U-MoE

activation peak dominates; extra collectives

PRINCIPLE

Cost-Aware Data Shard

equalize predicted work on every rank

equal total predicted cost

rank 0

rank 1

rank 2

rank 3

Shard

samples; fixed model on every rank

Comm

ordinary DDP after balanced assignment

U-MoE

use only with measured cost heterogeneity

NOT NOW

Tensor Parallel (TP)

split matrix dimensions within each layer

Weight matrix W

W0

W1

W2

W3

GPU 0

GPU 1

GPU 2

GPU 3

collective after layer GEMM

Shard

rows / columns of every large tensor

Comm

All-Reduce or All-Gather each layer

U-MoE

C=96 matrices too small for TP

NOT NOW

Pipeline Parallel (PP)

place consecutive layer stages on GPUs

GPU 0 Stage 0

GPU 1 Stage 1

GPU 2 Stage 2

GPU 3 Stage 3

m2

m1

m0

m2

m1

m0

m2

m1

m0

m2

m1

m0

P2P activations; micro-batch pipeline

Shard

contiguous layer stages

Comm

P2P activations and gradients

U-MoE

shallow backbone; bubble dominates

NOT NOW

Expert Parallel (EP)

place different experts on different GPUs

T0

T1

T2

T3

All-to-All token dispatch

GPU 0

E1-E3

GPU 1

E4-E6

GPU 2

E7-E9

GPU 3

E10-E12

Shard

expert weights and routed tokens

Comm

All-to-All dispatch + combine

U-MoE

12 small experts already fit locally

NOT NOW

Ulysses Parallelism (UP)

swap sequence shards for head shards

S/4 x H

S/4 x H

S/4 x H

S/4 x H

All-to-All: sequence > heads

S x H/4

S x H/4

S x H/4

S x H/4

local full-sequence attention; second A2A restores S/4

Shard

sequence first, then attention heads

Comm

two All-to-All operations per layer

U-MoE

window sequence is only 64 tokens

NOT NOW

Context Parallel (CP)

keep local Q; rotate K/V context blocks

GPU 0 Q0 + KV0

GPU 1 Q1 + KV1

GPU 2 Q2 + KV2

GPU 3 Q3 + KV3

Ring K/V online softmax

Shard

sequence context and K/V blocks

Comm

ring P2P K/V exchange

U-MoE

local 8 x 8 windows need no CP

Communication Legend

All-Reduce

Reduce-Scatter / All-Gather

All-to-All

P2P

Ring P2P

Measured Profile

4.11 M parameters; < 0.1 GB model states
61-76 GB peak is activation / loss / MoE capacity

Recommended for U-MoE-Fusion

DDP + bucket view + static graph + 8 MiB buckets + fused Adam
Cost-aware data assignment only after measured heterogeneity

Why Other Axes Wait

No parameter, expert-capacity, depth, or long-context bottleneck
Extra collectives would target the wrong resource